

Complexity-Guided Adaptive Diffusion Steps for Efficient Image Restoration

Harsha Basavaraj Beth
Boston University
Boston, MA, US
hbbeth@bu.edu

Abstract

Diffusion models have proven to be substantial in generative and image restoration tasks, yet their iterative sampling process remains to be a significant challenge for both time and resource-limited platforms. The critical issue noticeable in every related work is that the process has a fixed number of sampling steps. In this work, we investigate whether the number of diffusion steps can be adaptively selected by estimating image complexity. The term "image complexity" has several ways of being measured, such as, entropy, edge detection, image histogram distribution, human rating, etc. However, these metrics measure isolated aspects of an image rather than a holistic representation of image statistics. In contrast, a Gaussian Mixture Model (GMM), trained on natural image patches, captures a variety of image characteristics without being limited to a predefined notion of complexity. Accordingly, we use the negative log likelihood of the local image patches as our complexity metric. Given an effective complexity metric, we may then use a regression approach to adaptively predict the sampling steps required per image. The Denoising Diffusion Restoration Model (DDRM) is our baseline model which we evaluate deblurring on diverse images and compute correlations between complexity and required steps. Our results show a moderate correlation (r0.35) between GMM-based patch NLL features and required diffusion steps across a 100 ImageNet images, while a smaller GoPro deblurring subset showed stronger correlations for high-NLL patch statistics (r0.58). These results suggest that image complexity provides a useful signal for estimating diffusion step requirements, but simple statistical models capture only part of the underlying restoration difficulty. The code is available at: <https://github.com/HarshaBeth/Sample-Adaptive-Diffusion-Steps>.

1. Introduction

Diffusion models have advanced to be highly effective in performing image restoration tasks, such as denoising, de-

blurring, super-resolution, and artifact removal. The core is an iterative process of denoising a degraded image incrementally to achieve a high-quality generative result. However, the iterative nature of diffusion models is precisely what makes it a gradual process and computationally expensive. This poses a bottleneck for the adoption of more downstream tasks, which would not normally require more computation than needed, making inference time inefficient.

To resolve this bottleneck, existing methods propose reducing the number of steps based on mathematical manipulations that allow such reductions with an acceptable distortion-quality tradeoff. Nonetheless, some images may still not require the reduced number of diffusion steps. Images have varying complexity; a simple image requires few diffusion steps to achieve the same quantitative and qualitative results as a difficult image would with more steps. Therefore, a uniform step budget is suboptimal.

Such optimization methods exist to resolve the expensive steps constraint; DPM-solver proposes a faster solver to achieve competitive results with roughly 10 steps. Consistency models are defined by their training nature to map any noisy image along its diffusion trajectory to its final, denoised image, thus enforcing a "self-consistency" property. The self-consistency property enables it to directly predict the final image, removing the need for iterative solvers. The paper "Fewer Denoising Steps or Cheaper Per-Step Inference: Towards Compute-Optimal Diffusion Model Deployment" takes a unique approach; it focuses on compute optimization rather than step reduction. All things considered, the existing methods optimize globally and not individual input complexity. Thus, none of them focus on using steps only as required.

To address this limitation, I propose using image complexity to predict diffusion steps at the beginning of inference. Although images at time T are degraded, these images still carry semantic information respective to their ground truth counterparts. There are several methods to calculate "image complexity," such as entropy, edge density, etc. However, such metrics calculate only isolated parts of

complexity. Therefore, I propose to use a Gaussian Mixtures Model (GMM) to measure the complexity, as it is a probabilistic measure that models the full distribution of natural image patches. Natural images are a mix of distributions and a GMM captures several characteristics using its multiple components. GMMs are better used on patches of an image rather than the entire image; by taking image patches, we calculate the negative log likelihoods (the local complexity) and aggregate these values to capture the global complexity score. Then, we use a regression model to map the aggregated negative log likelihoods to the number of steps required for an image to reach a benchmark and compare our results. We observe that GMM finds irregular and rare structures, which is exactly what diffusion models struggle with; the negative log likelihood produced by the GMM indicates exactly that. All combined, it becomes our sample-adaptive inference pipeline to predict the number of steps.

2. Related Work

Diffusion Models. Denoising Diffusion Probabilistic Models (DDPM) [2] have emerged as a powerful approach for generative tasks, modeling the generative process by iteratively denoising images through 1,000 steps to produce high-quality image generation. Denoising Diffusion Implicit Models (DDIMs) [7] reduce the number of steps required to 50-100, following a deterministic, non-Markovian sampling procedure. Even with highly reduced steps, DDIMs can generate high-quality images but trade creativity for resources. Despite their impressive generative quality, these methods typically rely on a fixed number of sampling steps, which either takes up more or less than the necessary computation.

Fast Sampling Methods. DPM-solver [4] applies exponential integrators to approximate solutions to diffusion ODEs; the linear part of the ODE is integrated exactly, whereas the non-linear part gets approximated numerically using a neural network. Thus, enabling high-quality image generation in as few as 10 to 20 steps. Other approaches, such as consistency models [8], aim to speed up the process by mapping noise along the denoising trajectory directly to the clean image, thus ensuring *consistency*. Consistency models provide SOTA results with one-step sampling. However, even the most aggressive step reduction methods require fixed-step sampling; they reduce steps globally, not per image.

Diffusion for restoration. Denoising Diffusion Restoration Model (DDRM) [3] modifies the DDIM [7] to enable restorative tasks. Standard DDPM/DDIM models are trained on Gaussian noise, which means they cannot perform restorative tasks like deblurring, super-resolution, and artifact removal. DDRM is an unsupervised and efficient method that uses pretrained, unconditional diffusion models

to fill in missing information systematically without task-specific fine-tuning [3]. DDRM solves the inverse problem in the spectral domain of the degradation operator H . By treating the restoration as a variational inference problem, it can produce high-quality results in as few as 20 steps [3]. Nonetheless, DDRM uses fixed step counts across images.

Image Priors / Complexity. Natural image statistics are exactly what generative and restoration models aim to learn. Zoran and Weiss [10] showed that Gaussian Mixture Model (GMM) patch priors can effectively capture meaningful natural image structure and generate visually plausible image samples. Their Expected Patch Log Likelihood (EPLL) framework also applies the learned GMM prior to image restoration tasks such as denoising, inpainting, and deblurring [10]. In their work, GMM patch priors were shown to be effective with around 200 mixture components, suggesting that a moderately sized mixture model can capture substantial natural image patch statistics. These works use GMM patch models to represent natural image structure, but they do not connect this representation to diffusion sampling efficiency.

There exists a gap in existing work: either methods treat step reduction globally or image complexity is modeled separately. To the best of our knowledge, no prior work studies adapting diffusion sampling steps based on image-specific complexity. In this work, I propose to bridge this gap by mapping image complexity to the optimal number of diffusion steps required for achieving high-quality restoration results.

3. Methodology

3.1. Experiment setup

This research analyzes the sample-adaptive diffusion step selection strategy for image restoration purposes, namely, Gaussian deblurring. The restoration backbone used in this project is Denoising Diffusion Restoration Models (DDRM) [3]. DDRM leverages the sampling approach developed for DDIM [7] but adapts it to solve the linear inverse problems, including deblurring, super-resolution, and inpainting. Hence, in the context of this paper, DDRM is preferred over the vanilla DDIM sampler since the initial image is degraded by blurring and not generated with Gaussian noise.

The core idea of the experiment revolves around the question of whether or not all images should be restored with the same number of diffusion sampling steps. Existing diffusion restoration methods assume that every restored image requires the identical number of sampling steps, despite the variety of textures and structures that could be present within. Moreover, some images would require more sampling steps for their restoration in comparison to other images, making this approach less efficient.

Consequently, a range of sampling budgets is applied to the DDRM model to observe how different numbers of sampling steps influence the restored image quality. PSNR and LPIPS [9] are used to evaluate the restored image quality with respect to the reference images in terms of pixel-based reconstruction quality and perceptual similarity. It is not the aim of this research to alter DDRM; it aims to investigate whether it is possible to introduce sampling steps prediction in DDRM.

3.2. Complexity Estimation

To estimate image complexity, we use a Gaussian Mixture Model (GMM) over local image patches. Simple measures such as entropy or edge density capture only isolated aspects of complexity, whereas a patch-based GMM can model richer natural image statistics through its mixture components. Prior work has shown that GMM patch priors are effective for modeling natural image patches and for supporting image restoration tasks such as denoising, inpainting, and deblurring [10].

In this work, the GMM is used only as a feature extractor, not as a generative model for restoration. Each image is divided into local patches, and the trained GMM assigns a likelihood to each patch. Patches that are less likely under the learned natural image patch distribution receive higher negative log-likelihood values, indicating that they are statistically more complex or less typical. For an image with patches $\{p_i\}_{i=1}^N$ and a trained GMM density p_θ , the patch-level negative log-likelihood is defined as:

$$\text{NLL}(p_i) = -\log p_\theta(p_i). \quad (1)$$

The image-level complexity score is then computed by aggregating the patch-level NLL values:

$$C(x) = \frac{1}{N} \sum_{i=1}^N \text{NLL}(p_i). \quad (2)$$

This produces a single scalar complexity estimate for each degraded input image, which is later used to study its relationship with the number of diffusion steps required for restoration.

3.3. Step-Quality Relationship

To determine whether different images require different sampling budgets, we evaluate DDRM using multiple fixed diffusion step counts. For each degraded image, DDRM is run with several candidate values of k , where k represents the number of reverse diffusion sampling steps. Each restored output is then compared against the corresponding sharp ground-truth image.

Restoration quality is measured using PSNR and LPIPS [9]. PSNR captures pixel-level reconstruction accuracy, while LPIPS captures perceptual similarity using deep

visual features. Since this project focuses on paired image restoration rather than unconditional generation, FID and KID are not used as the primary metrics.

For each image, we define the best observed restoration quality across the tested step counts. The target step count k^* is then selected as the smallest number of diffusion steps that achieves quality close to the best observed result for that image. This allows the experiment to measure whether an image can be restored with fewer steps without losing significant reconstruction quality.

Formally, for an image x , let $Q(x, k)$ represent the restoration quality obtained using k diffusion steps. The optimal adaptive step count is defined as:

$$k^*(x) = \min \{k : Q(x, k) \geq Q_{\text{best}}(x) - \epsilon\}, \quad (3)$$

where $Q_{\text{best}}(x)$ is the best quality achieved among all tested step counts, and ϵ is a tolerance parameter that controls how close the selected result must be to the best observed quality. In practice, this produces a per-image target step count that can later be used for learning a relationship between image complexity and required diffusion steps.

3.4. Dataset Construction

The regression dataset is constructed in the form of input-label pairs $(C(x), k^*)$, where $C(x)$ represents the image complexity score and k^* represents the target number of diffusion steps. For each degraded image, the GMM-based negative log-likelihood features are computed and aggregated into a single image-level complexity value. This value serves as the input feature for the regression model.

The label k^* is obtained empirically by running DDRM on the same degraded image using multiple candidate step counts. Each restored output is compared against the corresponding sharp ground-truth image using PSNR and LPIPS. The selected k^* is the smallest step count that reaches restoration quality close to the best observed result for that image, according to the tolerance defined in the previous subsection.

Thus, each training example connects the statistical complexity of a degraded image to the number of diffusion steps required to restore it effectively. This dataset allows us to study whether GMM-based image complexity can serve as a useful predictor for adaptive diffusion sampling.

3.5. Regression Model

The regression model is used to predict the number of DDRM sampling steps required for a given degraded image. The input to the model is the GMM-based image complexity feature $C(x)$, and the target label is the empirically determined optimal step count k^* . The model learns a mapping:

$$\hat{k} = f_{\phi}(C(x)), \quad (4)$$

where f_{ϕ} is the regression function and $\hat{k}$ is the predicted number of sampling steps.

Because the dataset is small, we use a simple regression model rather than a large neural network. This helps avoid overfitting and allows us to directly analyze whether the GMM complexity score has a meaningful relationship with the required diffusion step count. The predicted value $\hat{k}$ is rounded to the nearest valid integer step count before being used during inference.

The regression model is trained to minimize the prediction error between the predicted step count $\hat{k}$ and the target step count k^* :

$$\mathcal{L}_{\text{reg}} = \frac{1}{M} \sum_{j=1}^M (\hat{k}_j - k_j^*)^2, \quad (5)$$

where M is the number of training samples.

3.6. Inference Pipeline

During inference, the input is a degraded image that must be restored. The image is first divided into local patches, and the trained GMM computes the negative log-likelihood of each patch. These patch-level values are aggregated to obtain the image-level complexity score $C(x)$.

The learned regression model then takes $C(x)$ as input and predicts the required number of DDRM sampling steps:

$$\hat{k} = f_{\phi}(C(x)). \quad (6)$$

The predicted step count $\hat{k}$ is rounded to the nearest valid integer step count and used as the sampling budget for DDRM. DDRM then performs restoration using the degraded image and the predicted number of reverse diffusion steps. The final output is the restored image.

This inference procedure allows the diffusion sampler to allocate computation adaptively: images predicted to be simpler can be restored with fewer steps, while images predicted to be more complex can be assigned more steps.

4. Experimental Setup

4.1. Datasets

We evaluate our approach on two datasets to capture a range of natural image characteristics. First, we use a subset of 100 images sampled from the ImageNet dataset [1], representing diverse scenes, objects, and textures. Second, we use a smaller subset of 22 images from the GoPro dataset [6], which contains real-world motion-blurred images and is commonly used for image deblurring tasks. The ImageNet subset provides diversity and scale, while the GoPro dataset allows evaluation on realistic blur patterns.

4.2. Degradation and Restoration

To simulate degraded inputs, we apply Gaussian blur to the clean images. Restoration is performed using the DDRM (Denoising Diffusion Restoration Models) framework [3]. We evaluate reconstruction quality across multiple diffusion step counts k , ranging from low to high values (e.g., 5 to 100 steps). For each image, we identify the minimum number of steps required to achieve a target reconstruction quality based on LPIPS [9], using a tolerance threshold relative to the best observed result.

4.3. Evaluation Metrics

Reconstruction quality is measured using LPIPS (Learned Perceptual Image Patch Similarity) [9], which captures perceptual similarity between reconstructed and ground-truth images. To evaluate the relationship between estimated complexity and required diffusion steps, we compute both Pearson and Spearman correlation coefficients. Pearson correlation measures linear relationships, while Spearman correlation captures monotonic trends, making it more robust to non-linearities in the data.

4.4. Complexity Estimation and Implementation Details

We estimate image complexity using a Gaussian Mixture Model (GMM) trained on local grayscale image patches. Patch-based GMM priors have been shown to effectively model natural image statistics and support restoration tasks [10]. The GMM is trained on grayscale patches extracted from a combined natural-image pool consisting of BSDS300 train/test images [5], sharp GoPro images [6], and the ImageNet subset used in our experiments [1]. Before patch extraction, images are resized to a fixed resolution of either 64×64 or 128×128 . Patches of size 8×8 are extracted with a stride of either 2 or 4 and flattened into feature vectors.

The GMM is trained with either 50 or 100 mixture components using diagonal covariance matrices. Input patches are standardized before training. For each degraded image, we compute patch-wise negative log-likelihood (NLL) values and derive summary statistics including mean, median, 90th percentile (p90), maximum, and top-10% mean NLL. These statistics serve as proxies for image complexity.

All experiments are conducted using consistent DDRM configurations [3], including fixed noise parameters and degradation settings, to ensure comparability across different diffusion step counts.

5. Results

5.1. ImageNet Results

We first evaluate the relationship between GMM-based complexity metrics and the required number of diffusion

steps on a subset of 100 images from ImageNet. The GMM is trained on a combined dataset of 3.3k images, resulting in approximately 742k extracted patches.

Across this dataset, we observe a moderate positive correlation between NLL-based complexity features and the required diffusion steps. In particular, mean NLL achieves a Pearson correlation of $r \approx 0.36$ and Spearman correlation of $r \approx 0.34$. Similarly, median NLL and p90 NLL show comparable performance, with Pearson correlations of $r \approx 0.33$ and $r \approx 0.36$, respectively.

These results indicate that global statistical measures of patch likelihood provide a consistent but limited signal for predicting diffusion step requirements. In contrast, max NLL and top-10% mean NLL exhibit slightly lower correlations on this dataset, suggesting that extreme local variations alone are not sufficient to explain restoration difficulty in highly diverse natural images.

Table 1. Correlation between GMM-based NLL features and required diffusion steps on the ImageNet subset.

Feature	Pearson r	Pearson p	Spearman ρ	Spearman p
Mean NLL	0.361	0.000226	0.342	0.000489
Std NLL	-0.007	0.942360	0.011	0.915998
Max NLL	0.278	0.005054	0.232	0.020142
Median NLL	0.330	0.000784	0.344	0.000465
P90 NLL	0.356	0.000283	0.318	0.001249
Top-10% Mean NLL	0.320	0.001178	0.266	0.007542

5.2. GoPro Results

We further evaluate the proposed approach on 22 sharp images from the GoPro dataset, where we synthetically apply a stronger Gaussian blur degradation. This setting provides a controlled test case with more severe blur than the ImageNet experiment.

On this dataset, we observe significantly stronger correlations for high-NLL-based features. In particular, max NLL achieves a Pearson correlation of $r \approx 0.58$ and Spearman correlation of $r \approx 0.64$, while top-10% mean NLL and p90 NLL achieve Pearson correlations of $r \approx 0.56$ and $r \approx 0.54$, respectively.

In contrast, global metrics such as mean NLL and median NLL show moderate but weaker correlations (Pearson $r \approx 0.40$). These findings suggest that, for structured degradation settings, the most complex local regions of an image play a dominant role in determining restoration difficulty.

However, due to the relatively small sample size, these results should be interpreted as preliminary but indicative.

Table 2. Correlation between GMM-based NLL features and required diffusion steps on the GoPro subset.

Feature	Pearson r	Pearson p	Spearman ρ	Spearman p
Mean NLL	0.399	0.065490	0.411	0.057564
Std NLL	0.055	0.808683	0.079	0.725676
Max NLL	0.579	0.004791	0.641	0.001296
Median NLL	0.397	0.067483	0.496	0.018802
P90 NLL	0.539	0.009644	0.496	0.018959
Top-10% Mean NLL	0.558	0.006906	0.614	0.002388

5.3. Feature Comparison

Across both datasets, we observe a clear distinction between global and local complexity measures. On the ImageNet dataset, global statistics such as mean and median NLL provide stable correlations, reflecting the overall diversity and variability of natural scenes. In contrast, on the GoPro dataset, features that emphasize the hardest local patches, such as max NLL and top-10% mean NLL, show substantially stronger predictive power.

These results suggest that diffusion step requirements are influenced by both global statistical complexity and localized difficulty, with the relative importance depending on the nature of the dataset. These results suggest that diffusion step requirements are influenced by both global statistical complexity and localized difficulty, with the relative importance depending on the nature of the dataset and degradation strength. In diverse, unconstrained datasets such as ImageNet, global complexity provides a more consistent signal, whereas in the GoPro sharp-image subset with stronger synthetic Gaussian blur, localized high-complexity regions appear to dominate the reconstruction process.

6. Analysis and Discussion

6.1. What Drives Diffusion Step Requirements

Our results suggest that the number of diffusion steps required for high-quality restoration is influenced by both global image statistics and localized regions of high complexity. On the ImageNet subset, global measures such as mean and median NLL show more stable correlations with k^* , indicating that overall image complexity provides a consistent signal for estimating restoration difficulty. However, on the GoPro subset, features that emphasize the most difficult patches, such as max NLL and top-10% mean NLL, show stronger correlations with the required number of diffusion steps.

This contrast suggests that diffusion restoration may be sensitive not only to the average complexity of an image, but also to concentrated difficult regions. Highly textured or structurally complex patches may require additional denoising iterations to recover details after degradation. In stronger Gaussian blur settings, these localized regions can dominate the restoration difficulty, whereas in more diverse natural-image settings, global complexity statistics appear

to provide a more reliable signal.

6.2. Why GMM-Based Complexity Works

The GMM-based approach captures image complexity by modeling the distribution of local patches and measuring how likely each patch is under this learned distribution. This allows the model to assign higher complexity scores to statistically rare or unusual regions, which often correspond to detailed textures, edges, or cluttered areas.

Unlike simple hand-crafted metrics such as entropy or edge density, the GMM provides a data-driven representation that captures multiple aspects of image structure simultaneously. The observed correlations indicate that this statistical notion of rarity aligns, to some extent, with the difficulty of diffusion-based reconstruction.

6.3. Limitations of GMM in Pixel Space

Despite capturing useful signals, the GMM-based approach has several limitations. First, it operates directly in pixel space, which lacks semantic understanding. As a result, it cannot distinguish between visually meaningful complexity (e.g., objects or scene structure) and low-level statistical variations. This limitation likely contributes to the moderate correlations observed on the ImageNet dataset.

Second, pixel-based patches are inherently noisy and highly correlated, making it difficult for the GMM to model their distribution effectively. This can reduce the reliability of NLL estimates, particularly for diverse and unconstrained datasets.

Finally, global statistics such as mean NLL may dilute the impact of highly complex local regions, while extreme-value metrics such as max NLL can be sensitive to noise or outliers. This highlights the challenge of designing robust summary features for complexity estimation.

6.4. Dataset-Dependent Behavior

The difference in results between ImageNet and GoPro highlights the importance of dataset characteristics. ImageNet contains highly diverse images with varying content, lighting, and structure, leading to more diffuse and less predictable relationships between complexity and restoration difficulty. In contrast, the GoPro dataset contains more structured motion blur, where localized high-complexity regions have a stronger and more consistent impact on reconstruction quality.

These findings suggest that the effectiveness of complexity-based step estimation may depend on the nature of the degradation and the underlying data distribution.

6.5. Implications for Adaptive Diffusion

Overall, our findings indicate that image complexity contains a meaningful signal for predicting diffusion step requirements, but simple statistical models capture only part

of the underlying factors. In particular, the importance of high-NLL regions suggests that adaptive diffusion strategies should account for localized difficulty, rather than relying solely on global image statistics.

This also motivates the use of more expressive feature representations, such as learned features from convolutional neural networks, which may better capture semantic and structural aspects of complexity and improve prediction accuracy.

7. Ablation Study

7.1. Effect of Patch Density

We first analyze the impact of patch density by varying the stride during patch extraction. Reducing the stride from 4 to 2 significantly increases the number of extracted patches (from approximately 45k to 168k) while keeping patch size fixed.

We observe a consistent improvement in correlation across most NLL-based features when using a denser patch sampling. For example, mean NLL correlation increases from $r \approx 0.35$ to $r \approx 0.36$, with similar gains for median and p90 NLL. This indicates that better coverage of local image regions improves complexity estimation, allowing the GMM to capture more fine-grained variations in texture and structure.

These results suggest that patch diversity and coverage are more important than simply increasing the number of images.

7.2. Effect of Image Resolution

Next, we evaluate the impact of increasing image resolution from 64×64 to 128×128 , while keeping patch size and stride fixed.

Despite the increase in total patches (to approximately 192k), we observe no significant improvement in correlation compared to lower-resolution inputs. In some cases, performance remains unchanged or slightly decreases.

This suggests that simply increasing global image resolution does not meaningfully improve complexity estimation, likely because the GMM operates on fixed-size local patches that already capture sufficient low-level structure. Instead, the quality of local sampling appears to be more critical than global scaling.

7.3. Effect of Training Dataset Size

We also analyze the effect of increasing the number of training images used for GMM fitting. Expanding the training set to approximately 3.3k images (resulting in over 700k patches) does not yield substantial improvements in correlation compared to smaller training sets.

This indicates that the model quickly reaches a representation ceiling, where adding more data does not signifi-

cantly improve the learned patch distribution. This further reinforces that model capacity, rather than dataset size, becomes the limiting factor.

7.4. Effect of GMM Capacity

To evaluate the role of model capacity, we increase the number of GMM components from 50 to 100 and 200.

Increasing the number of components to 100 leads to a notable improvement in correlation, with p90 NLL reaching approximately $r \approx 0.38$, representing the strongest performance observed in our experiments. This suggests that a higher-capacity model can better capture the diversity of patch distributions.

However, further increasing the number of components to 200 does not yield additional gains and may slightly degrade performance, indicating diminishing returns and potential overfitting.

7.5. Effect of LPIPS Threshold (ϵ)

We also study the sensitivity of the estimated optimal step count k^* to the LPIPS tolerance threshold ϵ . Recall that k^* is defined as the smallest number of diffusion steps that achieves restoration quality within a tolerance ϵ of the best observed LPIPS score for a given image.

On the GoPro subset with a stronger Gaussian blur, using a more relaxed tolerance threshold, such as $\epsilon = 0.3$, produces stronger correlations between k^* and high-NLL-based complexity features. In particular, max NLL and top-10% mean NLL show the strongest relationship with the required step count, reaching correlations of approximately $r \approx 0.58$.

In contrast, on the ImageNet subset, using an overly relaxed threshold, such as $\epsilon = 0.5$, leads to a noticeable drop in correlation. This suggests that when the tolerance is too loose, k^* becomes less informative because many images can be assigned small step counts even if their restoration difficulty differs.

Overall, these results indicate that the choice of ϵ should be aligned with the degradation severity and dataset characteristics. Moderate tolerance values provide a more reliable signal for connecting GMM-based complexity estimates $C(x)$ with the adaptive diffusion step target k^* .

8. Limitations

Despite demonstrating a measurable relationship between image complexity and diffusion step requirements, our approach has several limitations.

8.1. Coarse Estimation of Optimal Steps

The ground-truth adaptive step count k^* is approximated using a discrete sweep over a fixed set of candidate step values, followed by threshold-based selection using LPIPS.

This introduces quantization error, since the true optimal number of steps may lie between the evaluated values. While this approximation is sufficient for a proof-of-concept study, it limits the precision of the correlation analysis.

8.2. Limited Dataset Size

Our experiments are conducted on relatively small subsets, consisting of 100 ImageNet images and 22 GoPro images. While these subsets provide useful initial insights, the limited sample size, particularly for the GoPro subset, restricts the statistical reliability and generalizability of the observed correlations.

8.3. Pixel-Space GMM Representation

The GMM operates directly on raw grayscale image patches, which limits its ability to capture semantic information. As a result, the model captures low-level statistical variations but cannot account for higher-level structures such as object identity, scene layout, or contextual relationships. This likely contributes to the moderate correlations observed, particularly on diverse datasets such as ImageNet.

8.4. Sensitivity to Hyperparameters

The results are sensitive to several design choices, including patch size, stride, number of GMM components, and the LPIPS threshold. While we perform ablations to study these factors, selecting optimal configurations remains non-trivial and may vary across datasets and degradation settings.

8.5. Simplified Degradation Model

For the GoPro experiments, we use synthetically applied Gaussian blur on sharp images rather than the original motion-blurred inputs. This creates a controlled setting but does not fully reflect real-world degradation complexity. As a result, the findings may not directly generalize to more realistic restoration scenarios.

8.6. Human Evaluation Limitations

We include a preliminary comparison with human-assigned complexity scores based on annotations from three individuals. However, the overall human evaluation results were not strong and showed only limited agreement with the required diffusion step counts. This suggests that visually judging restoration difficulty is itself subjective and unreliable. Additionally, the small number of annotators limits the strength of any conclusions drawn from the human study.

9. Future Work

While our results demonstrate that image complexity contains useful signal for estimating diffusion step require-

ments, several directions can improve this approach.

Learned Feature Representations. A key limitation of the current method is the use of pixel-space patches. Future work can explore extracting features from pretrained convolutional neural networks and modeling their distributions instead. Such feature spaces capture higher-level structural and semantic information, which may better align with restoration difficulty.

Improved Estimation of Optimal Steps. The current estimation of k^* is based on coarse discrete sweeps and thresholding. A more precise approach, such as binary search over diffusion steps, could provide more accurate ground-truth labels and improve the reliability of the correlation analysis.

Larger and More Diverse Datasets. Expanding experiments to larger datasets would improve statistical reliability and help validate whether the observed trends generalize across different image domains and degradation settings.

Learned Step Prediction Models. Instead of relying solely on correlation analysis, future work could train regression or classification models to directly predict the required number of diffusion steps from complexity features. This would enable a fully automated adaptive diffusion pipeline.

Extension to Real-World Degradations. Our current experiments rely on a synthetic Gaussian blur for controlled evaluation. Future work should evaluate the approach on real-world degradations, such as motion blur and noise, to assess its practical applicability.

10. Conclusion

In this work, we investigated whether image complexity can be used to adaptively estimate the number of diffusion steps required for image restoration. We proposed a GMM-based approach that models local image patches and derives complexity metrics using negative log-likelihood (NLL).

Our experiments on ImageNet and GoPro subsets show that complexity features exhibit a moderate but consistent correlation with the required diffusion steps. In particular, global statistics such as mean and median NLL provide stable signals, while high-NLL features capture the influence of difficult local regions. However, the results also highlight the limitations of simple statistical models, particularly in capturing higher-level semantic structure.

Overall, our findings suggest that while image complexity is a promising direction for adaptive diffusion, more expressive representations and more precise estimation methods are necessary to achieve accurate step prediction. This work provides a foundation for future research on efficient and adaptive diffusion-based restoration.

References

- [1] Jia Deng, Wei Dong, Richard Socher, Li-Jia Li, Kai Li, and Li Fei-Fei. Imagenet: A large-scale hierarchical image database. In *CVPR*, pages 248–255, 2009.
- [2] Jonathan Ho, Ajay Jain, and Pieter Abbeel. Denoising diffusion probabilistic models. In *NeurIPS*, pages 6840–6851, 2020.
- [3] Bahjat Kawar, Michael Elad, Stefano Ermon, and Jiaming Song. Denoising diffusion restoration models. In *NeurIPS*, 2022.
- [4] Cheng Lu, Yuhao Zhou, Fan Bao, Jianfei Chen, Chongxuan Li, and Jun Zhu. Dpm-solver: A fast ode solver for diffusion probabilistic model sampling in around 10 steps. In *NeurIPS*, 2022.
- [5] David Martin, Charless Fowlkes, Doron Tal, and Jitendra Malik. A database of human segmented natural images and its application to evaluating segmentation algorithms and measuring ecological statistics. In *ICCV*, pages 416–423, 2001.
- [6] Seungjun Nah, Tae Hyun Kim, and Kyoung Mu Lee. Deep multi-scale convolutional neural network for dynamic scene deblurring. In *CVPR*, pages 3883–3891, 2017.
- [7] Jiaming Song, Chenlin Meng, and Stefano Ermon. Denoising diffusion implicit models. In *ICLR*, 2021.
- [8] Yang Song, Prafulla Dhariwal, Mark Chen, and Ilya Sutskever. Consistency models. In *ICML*, 2023.
- [9] Richard Zhang, Phillip Isola, Alexei A. Efros, Eli Shechtman, and Oliver Wang. The unreasonable effectiveness of deep features as a perceptual metric. In *CVPR*, pages 586–595, 2018.
- [10] Daniel Zoran and Yair Weiss. From learning models of natural image patches to whole image restoration. In *ICCV*, pages 479–486, 2011.